Sree Gnanesh Dommeti

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Education

University of Maryland, College Park **May 2026**
M.S. in Applied Machine Learning; GPA: 3.88 / 4.0 *College Park, MD*

National Institute of Technology, Calicut **May 2022**
B.Tech. in Computer Science and Engineering; GPA: 8.21 / 10 *Calicut, India*

Experience

GAMMA Lab, University of Maryland **January 2026 – Present**
Research Assistant, Advisor: Dinesh Manocha *College Park, MD*

- **TraceGround: Grounded Reasoning Evaluation for MLLMs** – researching methods that decompose multimodal model outputs into reasoning steps and check each step against visual/textual evidence.
- Developing trace-level checks to expose unsupported intermediate claims, hallucinated visual references, and weak evidence chains in multimodal responses.

FROOT Lab, University of Maryland **November 2025 – Present**
Research Assistant, Advisor: Alan Zaoxing Liu *College Park, MD*

- Contributing to ProjectASAP, an approximation-first telemetry research effort that accelerates SQL/PromQL analytics for agent-era data pipelines.
- Built PromQL workload-analysis, sketch-planning, and streaming-summary components across Prometheus `remote_write`, Arroyo, Kafka, and Rust query-serving paths.
- Additional NetArena work: built a Dockerized MALT purple agent for data-center capacity-planning benchmarks with LiteLLM prompting, output guards, retry logic, and correctness/safety/latency tracking.

Qualcomm **July 2022 – July 2024**
Software Engineer *Hyderabad, India*

- Maintained and debugged LTE Layer 3 RRC modules including Cell Selection, Paging, Security, Handover, and System Information for 4G/5G, CAT-M1, and NB-IoT modem stacks.
- Built Python log-analysis and regression-triage tooling for changelist assignment, RRC ownership routing, and diagnostic dashboards, reducing manual engineering effort by 40%.

Projects

TCOVITA - Temporally Consistent Video Instance Segmentation | *PyTorch, Mask2Former, VITA, InfoNCE*

- Extended VITA's object-token paradigm with a Temporal Association Module using Hungarian matching over cosine similarity and IoU, plus InfoNCE contrastive loss for stable cross-frame identity assignment.
- Implemented training and evaluation on a shared HPC cluster; benchmarked on YouTube-VOS 2019 with 3,471 train / 507 validation videos and 65 object categories.
- Reduced ID switches by 32.8%, improved Association Accuracy by 41.4%, and decreased track fragmentation by 42.1% versus VITA while maintaining segmentation quality (J=0.63, F=0.61).

DiffSplat: 3D Gaussian Generation and Editing | *PyTorch, CUDA Rasterization, 3DGS, SAM, Inpaint360GS*

- Set up DiffSplat text-to-3D Gaussian generation with released checkpoints, compiled the CUDA rasterizer, and patched renderer / multi-view attention compatibility in A100 environments.
- Ran CLIP-based ablations across SD 1.5, PixArt-Sigma, SD 3.5 M, guidance scale, seed sensitivity, prompt category, and native 3D versus independent 2D views.
- Built a SAM 3 + Inpaint360GS object-removal extension with Gaussian pruning, virtual-view inpainting, and depth repair; reached masked PSNR 32.77, SSIM 0.9855, LPIPS 0.0082 at 12K iterations.

Person Re-Identification on Market-1501 | *PyTorch, ResNet-50, BNNeck, Metric Learning*

- Implemented a strong ReID baseline with ResNet-50 + BNNeck, label-smoothed cross-entropy, triplet loss, PK sampling, random erasing, AdamW, warmup, and cosine decay.
- Achieved approximately 94% Rank-1 accuracy and 85% mAP on Market-1501 without re-ranking; analyzed failure modes under occlusion, similar clothing, and cross-camera domain shift.

Skills

ML / Research: PyTorch, TensorFlow, JAX, Scikit-learn, HuggingFace Transformers, Mask2Former, VITA, 3D Gaussian Splatting, SAM, CLIP, LoRA, DPO, RAGAS
Data / Systems: Python, SQL, C, C++, Rust, Kafka, Arroyo, Prometheus, Grafana, Elasticsearch, FAISS, Pinecone, PostgreSQL
MLOps / Infra: Docker, Kubernetes, AWS EC2/S3/SageMaker/ECS, MLflow, DVC, ClearML, W&B, GitHub Actions, OpenTelemetry